PROS AND CONS OF DIFFERENT TIM FILLERS

Type of Filler	Pros	Cons
Alumina fillers	**Low cost:** relatively inexpensive, making them cost-effective **Low electrical conductivity:** suitable for electrically sensitive applications **High melting point and chemical inertness:** stable and durable in various environments	**Low thermal conductivity:** compared to boron nitride or carbon nanotubes **High surface area:** hinders flow and increases viscosity in large-scale applications **Abrasiveness:** can cause damage in specific applications **Agglomeration:** aluminium nanoparticles, if not mixed properly, can cause density issues and poor thermal performance
Aluminium Hydroxide (Al(OH)3)	**Lower cost:** offers a significant reduction in cost (around 20% or more) **Flame retardancy:** emits water vapour when heated, enhancing fire protection	**Low thermal conductivity:** less effective for heat dissipation than alumina
Aluminium Nitride (AlN)	**High thermal conductivity:** suitable for applications requiring efficient heat dissipation	**High cost:** limits its use in cost-sensitive applications
Magnesium Oxide (MgO)	**Good thermal conductivity:** relatively inexpensive **Non-toxic and dielectric:** beneficial for various applications	**Special treatment required:** requires treatment technologies, complicating manufacturing, and processing
Zinc Oxide (ZnO)	**Fine particle size:** advantageous for specific applications	**Toxicity and regulatory issues:** classified as toxic in Europe, leading to potential regulatory limitations

will remain in focus as electronic performance and efficiency limits are pushed further. These materials facilitate effective heat transfer between heat sources and heat sinks, applied outside the package, typically between the package and the heatsink. They help dissipate heat across the entire package.

TIM 1 includes die attach materials and substrate attach materials, critical for ensuring thermal conductivity at the die level. Due to their superior thermal conductivity and performance characteristics, these materials are typically more expensive than other types of TIM. TIM 1 materials connect the die to the heat spreader, ensuring direct chip heat dissipation.

TIM 2 refers to the thermal interface materials used between the base plate and the heatsink in power electronics assemblies. While TIM 2 is cheaper per unit than TIM 1, its market value is higher due to larger usage in power electronics.

The effectiveness of TIMs hinges on their fillers, which can be ceramic materials like alumina, zinc oxide, or boron nitride. The choice of filler impacts several key properties:

Thermal conductivity. Essential for efficient heat transfer.

Mechanical properties. Affecting viscosity and application ease.

Cost. More expensive fillers translate to higher overall costs.

Common TIM forms include gap fillers, thermal greases, phase-change materials, adhesive tapes, and liquid metals. Each form offers unique advantages tailored to specific applications. In the EV sector, thermal management is crucial, particularly for battery technologies and power electronics. Silver and copper sintering technologies are emerging as significant players. While silver sintering is more prevalent, copper sintering offers a cost-effective alternative with comparable thermal conductivity. Major automakers, including Tesla and Volkswagen, are exploring advanced TIMs to improve battery performance and longevity.

Data centres demand robust thermal management to prevent overheating and ensure uninterrupted operation. TIMs play a vital role in maintaining optimal temperatures and enhancing the performance and reliability of servers and other critical infrastructure.

As autonomous vehicles become more sophisticated, the thermal management of advanced driver assistance systems (ADAS) components is crucial. TIMs manage heat in ADAS components, like sensors and processors, ensuring reliable operation in varied environments.

The deployment of 5G technology brings about higher power densities and increased thermal loads. TIMs manage heat in 5G infrastructure, ensuring reliable and efficient network operations.

TIM market set to soar

The market for thermal interface materials is set for significant growth, driven by advancements in electronics, electric vehicles (EVs), data centres, and 5G technology. Innovations in filler materials and formulations are expected to enhance performance and reduce costs, making TIMs more accessible across industries. In the expanding

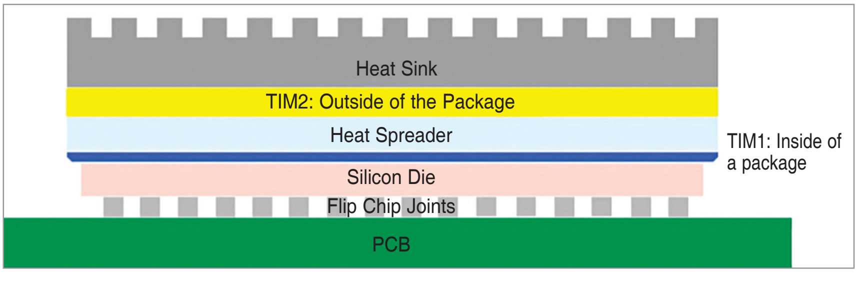

Fig. 1: Schematic to show the potential location of TIM in electronic devices